



POLITECNICO
MILANO 1863

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 3 – Euler Equations

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Task 1: Simulate the rotational motion of a 3U Cubesat

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.055 \text{ kg m}^2$$

$$I_z = 0.025 \text{ kg m}^2$$

Initial conditions

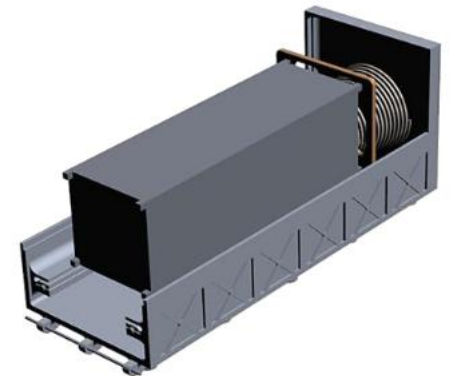
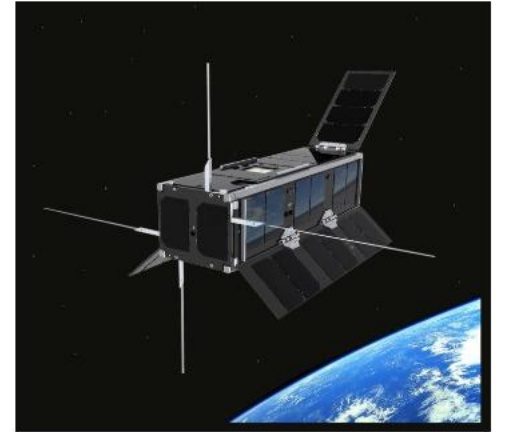
$$\omega_x(0) = 0.45 \text{ rad/s}$$

$$\omega_y(0) = 0.52 \text{ rad/s}$$

$$\omega_z(0) = 0.55 \text{ rad/s}$$

Governing Equations

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$



Tasks

- Use a scope to analyze the output
- Plot the output in MATLAB, with proper axis labels and units

Task 2: Verify the analytical solution for the symmetric case

Principal moments of Inertia

$$I_x = 0.0504 \text{ kg m}^2$$

$$I_y = 0.0504 \text{ kg m}^2$$

$$I_z = 0.0109 \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = \text{any rad/s}$$

$$\omega_y(0) = \text{any rad/s}$$

$$\omega_z(0) = \text{any rad/s}$$

Governing Equations

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$

Analytic Solution

$$\begin{cases} \omega_x(t) = \omega_x(0) \cos(\lambda t) - \omega_y(0) \sin(\lambda t) \\ \omega_y(t) = \omega_x(0) \sin(\lambda t) + \omega_y(0) \cos(\lambda t) \\ \omega_z(t) = \omega_z(0) \end{cases}$$

with

$$\lambda = \frac{I_z - I_x}{I_x} \omega_z$$

Question

Does the analytic solution provide a good approximation for the asymmetric case? Try with

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.055 \text{ kg m}^2$$

$$I_z = 0.025 \text{ kg m}^2$$

Task 3: Assess the stability of Euler Eqs equilibrium points

Principal moments of Inertia

$$I_x = 0.0250 \text{ kg m}^2$$

$$I_y = 0.055 \text{ kg m}^2$$

$$I_z = 0.0700 \text{ kg m}^2$$

Initial conditions

$$\omega_i(0) = 2\pi \text{ rad/s}$$

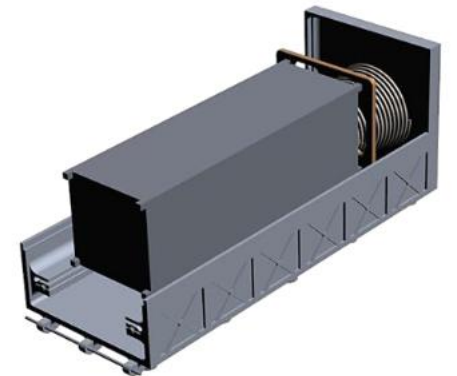
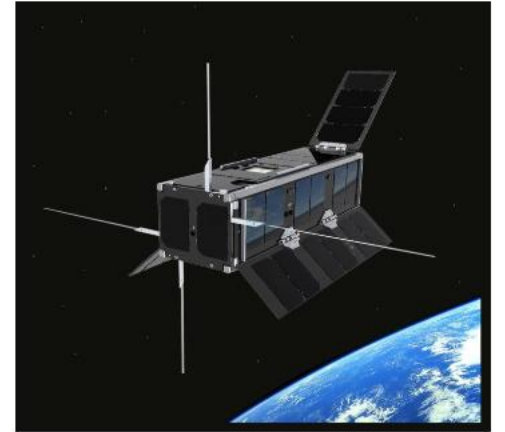
The subscript i denotes the spinning axis

Tasks

- Evaluate the stability switching the spinning axis direction and playing with different values of the off-axis initial angular velocities.

Governing Equations

$$\begin{cases} \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z \\ \dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_x \omega_z \\ \dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_y \omega_x \end{cases}$$



Task 4: Test the stability of a dual-spin spacecraft

Data

Principal moments of Inertia

$$I_x = 0.0700 \text{ kg m}^2$$

$$I_y = 0.055 \text{ kg m}^2$$

$$I_z = 0.025 \text{ kg m}^2$$

$$I_r = 0.0050 \text{ kg m}^2$$

Initial conditions

$$\omega_x(0) = 1 \cdot 10^{-6} \text{ rad/s}$$

$$\omega_y(0) = 1 \cdot 10^{-6} \text{ rad/s}$$

$$\omega_z(0) = 0.02 \text{ rad/s}$$

$$\omega_r(0) = 2\pi \text{ rad/s}$$

Governing Equations

$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z) \omega_z \omega_y - I_r \omega_r \omega_y \\ I_y \dot{\omega}_y = (I_z - I_x) \omega_x \omega_z + I_r \omega_r \omega_x \\ I_z \dot{\omega}_z = (I_x - I_y) \omega_y \omega_x - I_r \dot{\omega}_r \\ I_r \dot{\omega}_r = T_w = 0 \end{cases}$$

Tasks

- Evaluate the stability playing with different values of the off-axis initial angular velocities and wheel velocity
- Do the same with
$$I_z = 0.0504 \text{ kg m}^2$$
$$I_y = 0.0109 \text{ kg m}^2$$
- What happens in the two cases? Try to reason on your results.

